Real-Time Fraud Detection System with Explainable AI & Live Dashboard

Internship Capstone Project Summary

Dataset Overview

Dataset Used

IEEE-CIS Fraud Detection Dataset (Kaggle)

Files Used

- train_transaction.csv
- train_identity.csv

Dataset Statistics

- ~590,000 total transactions
- 433 merged features
- Severe class imbalance (~3.5% fraud rate)
- Datasets merged using TransactionID

Data Cleaning & Engineering

- Dropped columns with >50% missing values
- Applied median and mode imputation
- Encoded categorical variables using Label Encoding
- Applied RobustScaler for numerical stability
- Used SMOTE for fraud class balancing

Engineered Features

- AmtToMeanRatio
- HourOfDay
- DeviceRisk
- LogTransactionAmt
- IsNight
- EmailDomainMatch

Model Performance Summary

Three models were trained and evaluated on a stratified 80/20 split.

Model	PR-AUC	ROC-AUC	F1-Score
LightGBM (Tuned)	Best	Best	Best
XGBoost	2nd	2nd	2nd
Isolation Forest	Baseline	Baseline	Baseline

Best Performing Model — LightGBM

LightGBM outperformed all models due to:

- Better handling of imbalanced fraud data
- Faster training and inference
- Efficient histogram-based learning
- Higher Recall and PR-AUC
- Strong probability calibration for threshold optimization

Optuna/RandomizedSearchCV tuning further improved performance.

Threshold Optimization

Threshold vs F1-Score analysis identified the optimal fraud probability cutoff, significantly improving Recall compared to the default 0.5 threshold.

SHAP Explainability Findings

Top 3 Fraud Signals

1. Transaction Amount / AmtToMeanRatio

Fraudulent transactions showed unusually high or suspicious transaction amounts.

2. V-Feature Behavioral Patterns

Behavioral features strongly influenced fraud predictions and captured hidden transaction anomalies.

3. HourOfDay

Late-night transactions (midnight–4 AM) consistently showed higher fraud probability.

SHAP Waterfall Analysis

Confirmed Fraud Case (Probability ≥ 0.80)

High transaction amount, abnormal V-features, and risky device behavior strongly increased fraud probability.

Borderline Case (~0.50 Probability)

Mixed SHAP contributions resulted in moderate fraud confidence requiring secondary authentication.

Legitimate Transaction (Probability < 0.10)

Normal transaction patterns and low-risk features suppressed fraud probability.

Risk Segmentation Analysis

Transactions were segmented using fraud probability.

Risk Tier	Probability Range	Recommended Action
🔴 Critical Risk	≥ 0.75	Auto-block
🟡 Suspicious	0.40 – 0.74	OTP / MFA Verification
🟢 Clear	< 0.40	Auto-approve

Critical Risk Characteristics

Critical-risk transactions commonly showed:

- High transaction amounts
- Suspicious device usage
- Abnormal transaction timing
- Elevated fraud probability scores

- Behavioral anomalies

Business Recommendations

Policy 1 — Dynamic Risk-Based Blocking

Automatically block transactions classified as Critical Risk while routing Suspicious transactions to secondary authentication.

Benefits

- Reduces analyst workload
- Improves fraud capture rate
- Prevents high-risk transactions in real time

Policy 2 — Off-Hours Velocity Limits

Apply stricter transaction limits during midnight to 4 AM, the highest fraud activity window identified by SHAP.

Benefits

- Prevents large overnight fraud attempts
- Enables rapid customer verification
- Reduces financial exposure

Estimated Annual Savings

Assuming deployment at enterprise scale:

- Annual transaction volume: ~$50 Billion
- Estimated fraud exposure: ~$1.75 Billion
- Fraud recall at optimal threshold: ~85%
- Estimated fraud prevented: ~$1.49 Billion
- Investigation cost reduction: Significant operational savings

Estimated Net Annual Savings ~$1.48 Billion

Streamlit Dashboard Summary

An interactive multi-page Streamlit dashboard was developed.

Dashboard Features

Page 1 — Overview

- KPI cards
- Fraud rate analysis
- Risk tier donut chart
- Hourly fraud trend visualization
- Transaction amount distribution

Page 2 — Transaction Explorer

- Searchable transaction table
- Risk-based filtering
- Interactive scatter plots
- Transaction-level investigation

Page 3 — SHAP Explainer

- SHAP summary visualization
- Plain-English fraud explanation
- Transaction-level fraud reasoning

Technologies Used

- Streamlit
- Plotly
- SHAP
- Pandas
- LightGBM

Visualizations Created

- SHAP Summary Plot
- Fraud Probability by Hour

- Transaction Amount Histogram
- Risk Tier Donut Chart
- Precision-Recall Curve
- ROC Curve
- Threshold vs F1-Score Plot
- Fraud Risk Scatter Plot
- Correlation Heatmap
- Device Risk Distribution

Model Limitations

Despite strong performance, some limitations remain:

- Fraud patterns evolve over time (concept drift)
- False positives may impact genuine users
- SMOTE-generated synthetic samples may not cover all edge cases
- Lack of real-time streaming features
- Requires periodic retraining

Future Improvements

Additional data that could improve performance:

- Real-time geolocation
- Merchant category codes (MCC)
- Device fingerprinting
- Customer behavioral history
- IP address intelligence
- Fraud network graph analysis
- Streaming transaction velocity

Conclusion

This project successfully developed an end-to-end AI-powered fraud detection system combining:

- Advanced Machine Learning
- Explainable AI

- Fraud Risk Segmentation
- Interactive Analytics Dashboard
- Business-Oriented Fraud Insights

The final solution demonstrates real-world fintech deployment readiness and highlights the importance of explainable, data-driven fraud prevention systems in modern financial environments.